SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE OUTCOME COVERED
COURSE CODE: CSA15

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) Question Count: 5

Code of Conduct
Total Marks: 25

I Vijay A(192511030)(Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: Vijay A

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

1) Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet on demand, usually on a pay-as-you-go basis

Four characteristics are:

1) On-demand self-service: Users can access computing resources whenever required without directly contacting the service provider

2) Broad network access: Cloud services can be accessed through the internet using laptops, smartphone, and other devices.

3) Resource pooling: The provider shares computing resources among multiple users while keeping their data logically separate.

4) Rapid elasticity: Resources can be increased or decreased quickly according to demand.

Difference from traditional computing

Traditional on-premise computing requires organizations to purchase and maintain their own hardware, whereas cloud computing allows them to rent resources.

2) Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Cloud computing evolved through the following stages:

1) Mainframe computers: Large centralized computers served multiple users through terminals.

2) Personal computers: Computing became affordable and accessible to individuals and organization.

3) Client-server computing: Clients requested services from centralized servers over network.

4) Virtualization: A single physical server could run multiple virtual machines, improving hardware utilization.

5) Internet and web evolution: High-speed internet and web browsers enabled users to access applications remotely

6) Cloud computing: Virtualization, large data centers, automation, and internet connectivity enabled services such as AWS, Azure and Google cloud

3) Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Role of server virtualization

1) Resource utilization: Uses CPU, memory and storage efficiently

2) Cost reduction: Reduces the need for multiple physical servers

3) Isolation: Allows different OS and applications to run independently

4) Scalability: Makes it easier to create, modify and remove Vm

Difference between virtualization and cloud computing

Virtualization	cloud computing
→ Technology that creates virtual resources	→ Service model that delivers computing resources over the internet
→ Can operate within a private data center.	→ Provides on-demand access to computing services
→ Does not necessarily provide automated self-service or billing	→ Commonly includes self-service automation, usage-based pricing
Ex. Vmware ESXi	Ex: AWS, Azure, etc.

4) Discuss the role of data centers in delivering cloud services

Roles of data centers:

1) Computing: Provides servers to process applications and workloads
2) Storage: stores user files, databases, and application data.
3) Networking: Connects servers to the internet and enables communication
4) Availability: Uses redundant power supplies, cooling and network connections to reduce service interruptions
5) Security: Implements physical security, access controls, and monitoring to protect infrastructure
6) Scalability: Allows cloud provider to expand computing and storage capacity as demand increases.

5) Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

1) IaaS (Infrastructure as a service)

IaaS provides virtualized computing resources such as servers, storage, networks, and virtual machines over the internet. Users can install OS and applications according to their requirements. The cloud provider manages the physical hardware, while users manage the OS and applications

Ex: Amazon EC2 allows users to rent virtual servers for hosting websites and running applications

2) PaaS (Platform as a Service)

PaaS provides a ready-made platform for developing, testing and deploying applications. It includes development tools, runtime environments, and infrastructure management. Developers can focus on writing code without worrying about maintaining physical servers.

Ex: Google App Engine

3) SaaS (Software as a service)

SaaS provides ready-to-use software applications through the internet. Users can access the software using a web browser without installing or maintaining the application on their own computers. The service provider manages the software, updates, and infrastructure

Ex: Google docs

4) XaaS (Everything as a service)

XaaS is a broad cloud computing concept in which various IT resources, tools, and capabilities are delivered as services over the internet. It includes services over the internet.

Ex: Dropbox

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